Distractor Filtering Through Open Knowledge Graph Concept Proximity for Dense RAG Systems

Can Wikidata concept proximity filtering reduce distracting documents in dense retrieval and improve RAG accuracy compared to semantic-only baselines?

Retrieval-Augmented Generation (RAG) has become critical for grounding LLMs with external knowledge. However, recent work ("The Power of Noise") reveals that high-scoring retrieved documents may be distracting: semantically similar to the query but not containing the answer, leading to degraded LLM accuracy. Unlike state-of-the-art KG-based RAG systems, which leverage full graph semantics (relations, attributes), I propose using Wikidata concept proximity (graph topology) to identify and filter distractors. The hypothesis is that documents containing entities *closely connected* to query entities in the knowledge graph are less likely to be distractors than those with only semantic similarity.

Dataset: Natural Questions Open (NQ-open), filtered to queries where: (1) answers contain ≤5 tokens, and (2) both question and answer contain Wikidata entities recognized via **ReFiNed** (state-of-the-art end-to-end entity linker).

Pipeline:

1. **Corpus preparation**: Wikipedia dump (Dec 2018) segmented into 100-word passages avg. (sentence-boundary aligned), indexed via FAISS with Contriever embeddings.
2. **Query filtering**: Select NQ-open queries matching criteria (entity-linked Q&A, ≤5 token answers).
3. **KG subgraph construction**: Extract 3-hop neighborhood on Wikidata for all entities appearing in filtered queries and top-100 retrieved documents (via SPARQL API), store as local graph.
4. **Baseline Contriever-only**: Dense retrieval → top-5 documents → Llama2-7B generation.
5. **KG-Enhanced Reranking of top 100 dense retrieved documents:** use of kg_score = (connected_ratio) × (purity_ratio) where connected_ratio = (# query entities with ≥1 doc entity within 3-hop) / (total query entities) and where purity_ratio = (# doc entities within 3-hop of any query entity) / (total doc entities), documents are then reranked with final_score = α × contriever_score + (1-a) × kg_score with α + 0.5 and top-5 fed to Llama2-7B generation.
4. **Evaluation:** accuracy, measured as exact match following [1]. A response is considered correct if it contains at least one of the predefined correct answers from NQ-open (binary: correct/incorrect).

Expected Outcomes: KG proximity filtering will improve accuracy over semantic-only baseline by reducing distractors while preserving relevant documents.

[1] The Power of Noise: Redefining Retrieval for RAG Systems. SIGIR (2024)

[2] ReFinED: An Efficient Zero-shot-capable Approach to End-to-End Entity Linking. NAACL (2022)

[3] A Fair and In-Depth Evaluation of Existing End-to-End Entity Linking Systems (2023)

[4] A comprehensive Overview of Large Language Models (2024)

[5] Graph Retrieval-Augmented Generation: A Survey (2024)